Energy-selective four-time correlations in dissipative Majorana systems

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A known relaxation mode need not be visible in a correlation function, even when its decay rate is exact. We identify a four-time fermion correlation that selects energy relaxation in finite Majorana systems with homogeneous interactions and uniform linear loss and gain. Averaging each of two closely spaced fermion pairs over all flavors couples their leading nontrivial contribution to the Hamiltonian. After subtracting the product of the two within-pair correlations, equal pair durations ϵ give an energy signal of order ϵ^2, whereas the remaining contribution is of order ϵ^4 at fixed interpair delay. A finite-time bound controls this separation without requiring the full Liouvillian spectrum. Finite-system Sachdev–Ye–Kitaev calculations illustrate the resulting window and its decreasing signal strength. We also show that the two-point fermion correlator does not determine the complete finite-sample energy amplitude, and that mixing interaction orders destroys the leading selection of total energy. The results specify how an exact decay mode enters a concrete multitime correlation and when it can be isolated.

I. INTRODUCTION

Relaxation in an open quantum system depends on both its decay modes and the way they are excited and observed. The slowest Liouvillian mode can have little overlap with a chosen probe, while a faster mode can dominate the measured time window [1, 2]. Knowing an exact decay rate therefore leaves a physical question: which correlation reveals that rate, and how accurately can other contributions be excluded?

Majorana systems with linear dissipation offer a setting in which this question can be addressed at finite size. The dissipative Sachdev–Ye–Kitaev (SYK) model has been studied through its single-fermion Green's function and Liouvillian spectrum [3, 4]. For a Hamiltonian containing only q-body interactions, the energy is an exact decay mode with rate $q\mu$ [5], an example of the general relation between conserved quantities and simple dissipative decay [6]. This mode can be invisible to simple equal-time bilinears. It also need not be the slowest mode: at $q = 4, N = 6$, the parity-dressed energy has rate 2μ, below the energy rate 4μ [5].

Here we identify an energy-selective chronological four-time correlation. Two short fermion pairs, each averaged over every flavor, act as the source and readout of the intervening relaxation. Their leading nontrivial contribution selects the energy; finite pair durations introduce a quantitatively controlled admixture of other modes. The result uses neither a thermodynamic limit nor disorder averaging.

The ingredients have established precedents. Fermionic quantum regression fixes the signs of propagation between insertions [7], and multitime regression provides the general framework [8]. In closed, low-temperature SYK, nonalternating four-point correlations already admit an energy-vertex interpretation [9]. Our contribution is the finite-system dissipative selection rule, its finite-duration error, and the distinction between the complete energy amplitude and the information contained in a two-point correlator. All derivations are given in the Supplemental Material [10].

II. A CORRELATION THAT SELECTS ENERGY RELAXATION

Consider even N Majoranas, $\{\psi_i, \psi_j\} = \delta_{ij}$, with a nonzero Hamiltonian

$$H = \sum_{i_1<\cdots<i_q} J_{i_1\cdots i_q}\psi_{i_1}\cdots\psi_{i_q}, \qquad L_i = \sqrt{\mu}\,\psi_i. \tag{1}$$

We take real couplings, $q \equiv 0 \pmod 4$, and $\mu > 0$. Each jump exchanges a fermion with a Markovian environment; equal rates leave the maximally mixed state $\rho_\infty = I/d$, $d = 2^{N/2}$, stationary. In terms of complex fermions, a pair of Majorana jumps describes balanced loss and gain. The analytic results allow arbitrary couplings in Eq. (1). The numerical examples use the Gaussian SYK ensemble [9, 11], with $\overline{J_I^2} = J^2/\binom{N-1}{q-1}$.

Uniform noise damps a Majorana string according to the number of fermions it contains [12, 13]. Since every term in H has the same length, the exact energy autocorrelation is

$$\langle H(g)H(0)\rangle_\infty = ve^{-q\mu g}, \qquad v = d^{-1}\,\mathrm{tr}\,H^2. \tag{2}$$

This known result fixes the propagation after energy has been selected. The question here is how a sequence of elementary fermion insertions achieves that selection.

Take four ordered times with consecutive separations $s, g, u > 0$, as in Fig. 1(a). We study

$$G_4(s,g,u) = \frac{1}{N^2}\sum_{i,j}\big\langle\psi_i(s+g+u)\psi_i(s+g) \times \psi_j(s)\psi_j(0)\big\rangle_\infty. \tag{3}$$

The sum includes $i = j$. Subtracting the product of the two within-pair correlations gives $G_{\mathrm{pair}} = G_4 - C_N(u)C_N(s)$, where $C_N(t) = N^{-1}\sum_i\langle\psi_i(t)\psi_i(0)\rangle_\infty$.

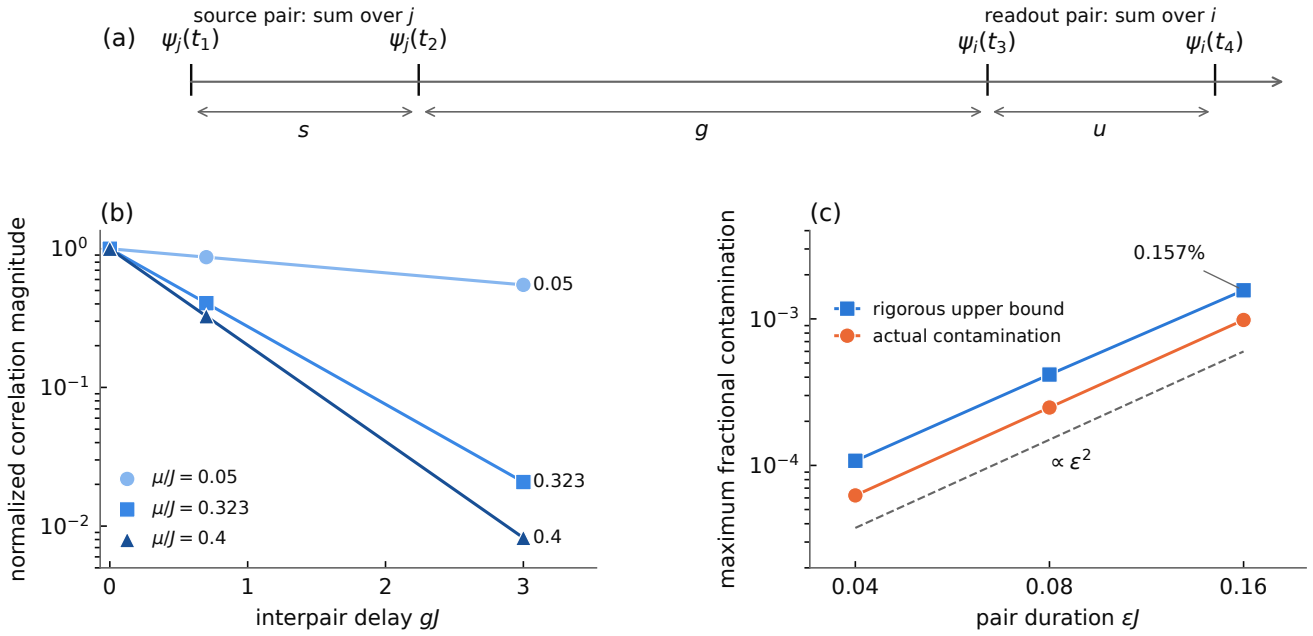


FIG. 1. Energy selection at finite pair duration. (a) Chronological insertions defining Eq. (3); the pair sums include every flavor combination. (b) Magnitude of $G_{\text{pair}}(\epsilon, g, \epsilon)$ normalized at $g = 0$, for one $N = 10, q = 4$ realization and $\epsilon J = 0.16$. Symbols are the three calculated delays; lines show $e^{-4\mu g}$ without a fitted rate. The middle noise value is $\mu/J = 0.3228079267$, rounded in the legend. (c) Largest relative contamination $|G_\perp|/|G_H|$ and largest rigorous bound among the 18 checked systems, at $gJ = 0.7$. The two maxima need not come from the same system. The dashed guide shows quadratic scaling, not a fitted exponent or a thermodynamic extrapolation.

This removes the stationary contribution between pairs; it is not the cumulant obtained by subtracting every Wick pairing. Fermionic regression is understood throughout. The observable is a multitime correlation, rather than the ordinary Hamiltonian linear susceptibility, which vanishes at ρ_∞.

The correlation separates exactly into an energy contribution and a remainder:

$$G_{\text{pair}}(s, g, u) = \mathcal{A}_H(s, u)e^{-q\mu g} + G_\perp(s, g, u). \quad (4)$$

The amplitude $\mathcal{A}_H$ is the product of the source and readout overlaps with energy. It can depend on μ through the dynamics within each pair; its dependence on the interpair delay is entirely the exponential in Eq. (4). Other modes, including ones degenerate with the energy rate, remain in $G_\perp$.

The selection mechanism follows from interaction homogeneity. Immediately after the first fermion insertion, the short-time change of the pair contains the interaction terms involving that flavor. Summing over all flavors counts every q-body interaction exactly q times. The result is proportional to the total Hamiltonian, with no component along another nonstationary direction. The second pair makes the same selection on readout. A single flavor instead probes a partial Hamiltonian, so the complete averaging in Eq. (3) is essential.

For equal short durations, $s = u = \epsilon$, the energy amplitude and the remainder obey

$$\mathcal{A}_H(\epsilon, \epsilon) = -\frac{q^2 v}{N^2}\epsilon^2 + O(\epsilon^3),$$
$$G_\perp(\epsilon, g, \epsilon) = O(\epsilon^4). \quad (5)$$

These are fixed-N, fixed-g statements. The negative leading amplitude is allowed for this ordered correlation; it is not a probability or an absolute square. The energy signal requires one short-time interaction in each pair, whereas an orthogonal contribution requires an additional order in both. Thus the relative contamination decreases quadratically as the pairs are shortened.

III. FINITE DURATION, SIGNAL STRENGTH, AND THE OBSERVATION WINDOW

Figure 1(b) illustrates the selected decay in a finite SYK sample. Agreement with the energy exponential concerns a particular correlation and time window; it makes no claim about which mode controls the full spectrum. The error can be bounded directly from the nonenergy components generated by the two pairs, without diagonalizing the complementary Liouvillian. The bound and its proof are given in Supplemental Material, Sec. V.

We evaluate the correlation and this bound for $q = 4$, $N = 6, 8, 10$, two realizations at each size, and three noise strengths $\mu/J = 0.05, 0.3228079267, 0.4$. At $gJ = 0.7$

and $\epsilon J = 0.16$, the relative-contamination bound is below 0.157% for all 18 systems; the largest actual contamination is 0.0983%. Figure 1(c) shows the improvement as the pair duration decreases. These maxima describe the checked systems, rather than a confidence interval over disorder. The small matrices also permit direct checks of the four separate insertions, the source–readout construction, and the analytic short-time coefficients.

Selection carries a signal cost. Equation (5) shows that halving the pair duration reduces the leading energy signal by a factor of four, along with the quadratic improvement in relative contamination. For the chosen SYK normalization, $\overline{v} = NJ^2/(q2^q)$, so the leading disorder-averaged signal at fixed g is $-qJ^2\epsilon^2 e^{-q\mu g}/(N2^q)$. This statement concerns the short-time coefficient; it does not assume finite-duration self-averaging. The present analysis establishes correlation selectivity, not a measurement advantage after sampling costs.

The interpair delay imposes a separate limitation. Energy decays at $q\mu$, while a complementary even mode can decay more slowly. Consequently, the relative-error bound can grow with g, even though its absolute upper bound decreases. Shortening the pairs opens a controlled finite-time window, but does not establish energy dominance at arbitrary late times. The small-ϵ expansion is also not uniform in system size without further estimates.

IV. AMPLITUDE INFORMATION AND THE ORIGIN OF SELECTIVITY

An exact decay rate does not fix its amplitude in Eq. (4). At finite pair duration, the source and readout have complex energy overlaps. Their imaginary parts are fixed by the time derivative of the two-point fermion correlator; their real parts generally survive in an individual sample. The leading real part is proportional to the third energy moment $d^{-1}\operatorname{tr} H^3$, which need not vanish even for a pure quartic Hamiltonian. Thus a two-point calculation alone does not determine the complete energy contribution.

Figure 2 quantifies the missing information for one $N = 8$ sample. At $\epsilon J = 1$, the complete energy amplitude is $-0.00838880 - 0.00051700i$, whereas keeping only the part fixed by the two-point function gives -0.00839676. The resulting complex-amplitude discrepancy is about 6.2%. This comparison concerns the isolated energy contribution, not the entire four-point function. A disorder ensemble symmetric under $H \mapsto -H$ removes the mean real vertex, but its covariance can still contribute to the mean energy amplitude. Replacing an average of products by a product of averages requires an additional self-averaging argument.

The selection rule also identifies its own failure mechanism. If $H = H_q + H_p$ contains two interaction orders, the first short-pair contribution weights them differently, coupling to $qH_q + pH_p$ rather than to total energy. The leading source therefore has a nonenergy component before the interpair evolution begins. In a quartic $N = 8$ sample with an added term $0.25J\, i\psi_0\psi_1$, this component accounts for 2.39% of the squared norm of the initial size-weighted vertex. Total energy is then generally no longer a single-rate decay mode. Homogeneous interaction order, together with uniform noise and the full flavor sum, is the physical condition behind the selectivity; randomness and chaotic dynamics are not required.

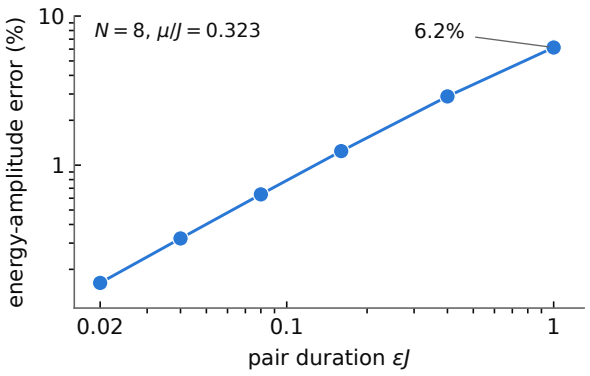


FIG. 2. Two-point information leaves part of the finite-sample energy amplitude undetermined. The plotted fractional discrepancy compares the complete $\mathcal{A}_H(\epsilon,\epsilon)$ with the result obtained by retaining only the imaginary energy vertices fixed by C_N. Data are for $N = 8, q = 4$, one realization, and $\mu/J = 0.3228079267$. Lines connect evaluated pair durations. This is an amplitude-reconstruction error within the energy contribution, distinct from the complementary-mode contamination in Fig. 1(c).

V. DISCUSSION

The four-time correlation specifies how the known energy decay of a dissipative Majorana system can be exposed without identifying the slowest complementary mode. Its finite-duration accuracy and its decreasing amplitude must be considered together. The incomplete two-point reconstruction further separates three pieces of information: the existence of a decay mode, its rate, and the source-dependent amplitude with which it enters a correlation.

This finite-system result complements the large-N collective description. General SYK four-point kernels and their sourced recursions are established [14, 15]. In the Supplemental Material we give an explicit energy eigenfunction of the continuous singlet kernel and relate its shape to the imaginary energy vertex. That relation supplies an analytic consistency check. It does not establish the complete finite-duration residue, spectral completeness, or a thermodynamic gap transition. In particular, loss of integrability in the relative-time representation does not remove the finite-system energy contribution.

The resulting benchmark is a concrete correlation with

a known energy component, a controlled finite-time remainder, and a specified breakdown under mixed interaction orders. It can be used to test approximations to multitime open-system dynamics. Extending it to a measurement protocol would require accounting for the decreasing signal and the cost of reconstructing the ordered correlation; those resources are not determined here.

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